
1.6 Clinical reasoning

This process involves the health worker being confident in their knowledge and skills, as well as knowing their limitations, and delivering the best care possible to the patient within the constraints of available diagnostic and therapeutic capacity and resources.

First, in every patient, triage for severe conditions and conditions that could potentially deteriorate quickly using the Quick Check (Section 2). Immediately provide emergency treatment and perform emergency laboratory investigations.

Thereafter, obtain more information about the presenting complaints and consider the signs and symptoms. Be sure to think again of serious or potentially life-threatening conditions associated with each symptom. Establish the possibility of such a condition, and keep it near the top of the list until safely excluded. Rapidly do relevant laboratory and other investigations for serious conditions. Initiate early investigations for serious conditions for which relevant tests are available at the health facility.

Next, ascertain the likely cause of each presenting symptom. Use the relevant differential diagnosis tables. This involves a process of weighing up the likelihood of one diagnosis over other possible diagnoses by gathering available evidence – history, physical examination, and further investigations. Consider:

- patient demographics – age, sex, pregnancy status
- risk factors – environmental factors and any others particular to the patient
- important negative findings – remember to actively look to exclude these
- combinations of signs and symptoms associated with a particular disease
- any history of prior intervention for the current condition.

Identify all diagnoses (more than one may be present). Plan treatment and consolidate a combined treatment plan, addressing the several problems an acutely ill patient may have. If there are many unexplained symptoms over time, consider the possibility of a mental health problem (see Section 10.11).

Clinical reasoning and medical uncertainty

Health workers in resource-limited settings frequently need to make clinical decisions with incomplete diagnostic support from radiology or the laboratory. The processes of clinical reasoning used, and the knowledge possessed to support decision-making, are critical determinants of the quality of clinical practice.

Clinical mentoring and supportive supervision are very important for good clinical decisions and for improving clinical practice over time. In areas with high levels of diagnostic and therapeutic capacity, poor decision-making wastes resources; a large proportion of interventions may be unnecessary while a large number of useful interventions may not be provided.

The **content** of clinical guidelines (such as lists of signs and symptoms, and treatment of common diseases) is very important. However, the **process** of clinical decision-making is somewhat distinct from these. Reaching an **evidence-based clinical decision** involves making a systematic health assessment of a patient based on history and physical examination, and linking this with information in the patient's medical records. Complete and accurate medical records on patients will enable the health worker to make better informed decisions.

Each diagnostic process begins with uncertainty but draws upon contextualized and case-specific knowledge, as well as increasingly on biomedical informatics and support tools. Clinicians transform the information or evidence available to them into a decision with consequent action, based on knowledge, the environmental, socioeconomic, and epidemiological context and the accumulated data on the specific case.

Clinical decision-making is centred on a **differential diagnosis** (abbreviated DDX throughout the manual). Initially, this should be broad, followed by progressive elimination of possibilities without sufficient evidence. This process of elimination includes both seeking evidence that supports a particular diagnosis and evidence to exclude a possibility. However, solely listing the conditions that could potentially account for the presenting symptoms in a patient is insufficient, especially in PLHIV. It is important to consider other serious diseases or co-morbidities that may be present. Consideration needs to be given to the possibility of disseminated disease affecting multiple organ systems, and diseases with diverse symptomatology (see Section 11). Appropriate context needs to be established by considering the patient's risk factors, as well as any unmet prevention needs.

The frequency and severity of a disease may influence how diseases within the differential diagnosis table are ranked, and the order in which they are investigated. Differential diagnosis (DDX) tables should be considered in the local context of diseases, both those that are endemic and epidemic in an area. Determining the immunological status of an HIV-infected patient may be useful for ranking the likelihood of a particular infectious agent. Additional or repeated physical examinations, laboratory tests, and other investigations, consultation with clinical mentors, and consideration of the local disease epidemiology, can assist in ruling in or out a diagnosis. It may be important to initiate early investigations for serious conditions for which relevant tests are available at the health facility (see Section 5.1).

If it is not possible to confirm a diagnosis at the facility, consider referral or the empirical treatment of common or life-threatening conditions, depending on local guidelines. As for all investigations and therapy, assessment of the risks is required, and of the benefit and cost of investigations versus empirical treatment. At regular intervals, it is necessary to revise an initial diagnosis and reassess clinical progress, particularly whether or not a patient is improving within the expected time frame.

Establishing clinical diagnosis using different differential diagnosis tables

1. Use the differential diagnosis tables to establish links between clinical features and possible underlying diagnoses.
2. Prioritize the list of possible diagnoses from the table based on the conditions most likely to exist in the setting or to be life threatening.
3. Request and perform specific diagnostic tests (such as lumbar puncture, skin scrapings, fine needle aspiration) in order to support or refute diagnoses from the initial differential list.
4. Identify patients who need hospitalization.
5. Determine whether clinical findings or diagnostic test results support a condition from the initial differential diagnosis list.
 - a. If yes, treat accordingly.
 - i. If treatment was successful, follow the patient as indicated.
 - ii. If treatment was unsuccessful, re-evaluate the patient, modify the differential diagnosis, and return to step 1.
 - b. If no, re-evaluate the patient, modify the differential diagnosis, and return to step 1.
6. If the diagnosis is uncertain:
 - a. Consider initiating empirical therapy for serious or life-threatening conditions.
 - b. Consider initiating empirical therapy for non-severe conditions when a diagnosis is likely and treatment is accessible and likely to be effective.

Improved clinical decision-making comes with experience and knowledge of local patterns of disease. For less experienced staff, supportive supervision and clinical mentoring are important in building confidence.

Avoiding errors in clinical reasoning

The following principles are often cited to guide the clinical reasoning process.

- Try to think of a single disease that accounts for most or all of the clinical findings ("Occam's razor"). This principle does not always apply in the elderly and in immunocompromised patients (e.g. patients with advanced HIV infection), where there may be more than one pathological process occurring at the same time, in the same or in different organs.
- Even if a clinical presentation looks similar to or is "representative of" a particular illness, this does not prove that the cause is due to that illness. Common diseases sometimes have uncommon presentations, and uncommon diseases can sometimes resemble those that are very common.
- An uncommon presentation of a common disease is generally more likely than a typical presentation of an uncommon disease. (Consider "Sutton's Law," named after a famous bank robber who explained that he robbed banks because "that's where the money is". This suggests that a clinician consider common causes in the local region for a patient's symptoms before considering uncommon causes.)
- Consider what could kill a patient quickly, even if the diagnosis may be uncommon (this counterbalances Sutton's Law).
- Plan the initial empirical or syndromic treatment so as to cover the most common causes and the most serious (life-threatening) possible causes.
- Avoid premature closure of the diagnostic process. Start with a broad differential diagnosis so as not to prematurely eliminate possibilities without sufficient evidence.
- Do not be overconfident. Seek reasons why decisions may be wrong and consider alternative hypotheses. Ask questions that would disprove, as well as prove the current hypothesis.
- Conditions recently seen can be over-diagnosed, especially those that were particularly dramatic, or in which a mistake was made that needs to be avoided in the future.
- Avoid "illusory correlation". This means that just because two findings occur together, it does not necessarily mean that one caused the other.
- Know what you do not know. If you have a knowledge gap, admit it and seek the missing information, e.g. from a book, from your colleagues and co-workers, a clinical mentor, from a warm-line (a phone consultation service that calls users back within a short period of time with relevant information and assistance), or from reputable internet sites.

5. Approach to laboratory investigations

5.1 Interpreting laboratory results

Evidence-based medicine: steps to using laboratory results

After taking a pertinent history and performing a physical examination, use your knowledge and the appropriate differential diagnosis tables to develop a relevant differential diagnosis, ranked both by what can be common causes and by what can be life-threatening causes.

- Laboratory tests are useful to confirm or rule out a diagnosis (or differential diagnoses); to establish the severity of disease (e.g. CD4 cell count); to monitor treatment outcomes; or to screen for disease (active TB case finding). The tests you choose to order are based on evidence-based health care, national guidelines, and your clinical judgement.
- Order the “best tests” you have available in your setting to either “rule in” or “rule out” a diagnosis that you are considering. Very few tests in medicine are perfect, so it is important that, as the clinician you know how accurate a test is before interpreting a result. For example, how accurate is a single expectorated sputum to diagnose pulmonary tuberculosis in someone with a lung cavity? How accurate is this test in someone without a lung cavity?
- The accuracy of a test can be described by its sensitivity, specificity, and predictive value:
 - **Sensitivity** refers to the ability of the test to correctly identify individuals who truly have the disease. If you perform a test that is highly sensitive for a particular disease and the result is negative, it is very unlikely that that disease is present; hence, the test has been helpful in ruling out the disease in question.

Example: the malaria Rapid Diagnostic Test (RDT) is a very sensitive test. Therefore, if the result is negative, the possibility of malaria has been ruled out. The patient does not have malaria.

- **Specificity** refers to the ability of the test to correctly identify individuals who do not have the disease. If you perform a test that is highly specific for a given disease and the result is positive, you can now be more certain that you have made the correct diagnosis; hence, the test has been helpful in ruling in the disease in question.

Example: an AFB smear on CSF is a very specific test. Therefore, if the result is positive, the possibility of tuberculous meningitis has been ruled in. The patient has tuberculous meningitis.

- **The predictive value** of a test (also called the post-test probability of disease) refers to the ability of the test to correctly identify the disease. Unlike sensitivity and specificity, which do not vary within populations, the predictive value of a test depends on age, gender, geographic location, and disease prevalence.

Test your knowledge of evidence-based decision-making by considering a clinical case.

- A 36-year-old man started ART (AZT + 3TC + EFV) in April.
- His pre-treatment CD4 was 15. He is at WHO clinical stage 3, with oral thrush.
- In June, two months after starting ART, he presented with severe headache, confusion, a stiff neck, and fever.
- His chest X-ray was normal.
- The CSF indicated:
 - 19 polys, 253 lymphs
 - protein 0.92
 - glucose 2.6
 - Gram stain – no bacteria

Question:

- What is your differential diagnosis for meningitis?

Differential diagnosis:

- Tuberculous meningitis
- Cryptococcal meningitis
- Bacterial meningitis (partially treated)
- Lymphomatous meningitis

You decide to perform an AFB smear on the CSF. What is the probability that the meningitis of this patient is due to tuberculosis 1) if the test is positive? 2) if the test is negative?

These probabilities depend on the sensitivity and specificity of the test, as described above, and also on how frequent the disease is in your region (prevalence of the disease in the general sick population, also called “pre-test probability”, as it is the probability that the patient has the disease before any testing).

Situation A

Let us say that evaluation of a cohort of AIDS patients living in your region has shown that 20% of meningitis is due to tuberculosis. You can draw the following 2-by-2 table:

Step 1: Among 1000 patients, 200 (20%) have the disease and 800 do not have the disease.

		Tuberculous meningitis		Total patients
		Present	Absent	
Result of AFB smear of CSF	Positive			
	Negative			
Total patients		200	800	1000

Step 2: The **sensitivity** of AFB smear on CSF is 60%. Thus, among 200 patients having the disease, 120 tests (60%) will be positive.

The **specificity** of AFB smear on CSF is 99%. Thus, among 800 patients not having the disease, 792 tests (99%) will be negative.

		Tuberculous meningitis		
		Present	Absent	Total patients
Result of AFB smear of CSF	Positive	120	8	120 + 8 = 128
	Negative	80	792	80 + 792 = 872
	Total patients	200	800	1000

$\Downarrow$
 $\frac{120}{200}$
 $=$
 sensitivity = 60%

$\Downarrow$
 $\frac{792}{800}$
 $=$
 specificity = 99%

- Step 3:
- (a) The Positive Predictive Value (PPV) is $120/128 = 0.94$. Thus, if the AFB smear on CSF is positive, the (post-test) probability that the patient has tuberculous meningitis is 94%.
- (b) The Negative Predictive Value (NPV) is $792/872 = 0.91$. Thus, if the AFB smear on CSF is negative, the (post-test) probability that the patient actually has tuberculous meningitis is only 9% (100%–91%).

		Tuberculous meningitis		
		Present	Absent	Total patients
Result of AFB smear of CSF	Positive	120	8	128
	Negative	80	792	872
	Total patients	200	800	1000

$\Rightarrow 120/128 = 94\% = \text{PPV}$

$\Rightarrow 792/872 = 91\% = \text{NPV}$

Situation B

If the cohort of AIDS patients living in your region has shown that in fact only 2% of meningitis is due to tuberculosis, the 2-by-2 table will change in the following way:

		Tuberculous meningitis		Total patients	
		Present	Absent		
Result of AFB smear of CSF	Positive	12	9	21	⇒ 12/21 = 57% = PPV
	Negative	8	871	879	⇒ 871/879 = 99% = NPV
	Total patients	20	880	1000	
		↓	↓		
		$\frac{12}{20}$	$\frac{871}{880}$		
		=	=		
		sensitivity = 60%	specificity = 99%		

In this situation:

- If the AFB smear on CSF is positive, the (post-test) probability that the patient has tuberculous meningitis is only 57%. Hence, the etiology of the meningitis might be tuberculosis, but it might also be a disease other than tuberculosis. Further investigations are necessary.
- If the AFB smear on CSF is negative, the (post-test) probability that the patient has tuberculous meningitis is only 1% (100–99%). The possibility of tuberculous meningitis is thus fully excluded.

Table: Sensitivity and specificity for selected diagnostic tests			
Disease	Test	Sensitivity	Specificity
HIV	HIV ELISA	100%	98%
HIV	HIV rapid tests	99%	98%
Malaria	Malaria smear	52.5%	77%
Syphilis	RPR/VDRL	91%	95%
	FTA-ABS	92%	96%
Pulmonary tuberculosis – culture positive	3 expectorated sputum smears ¹	70%	96%
	Antibiotic trial to rule out pulmonary TB in smear negative ²	55%	77%
Cryptococcal meningitis	CSF India ink ³	72.6%	99%
	CSF cryptococcal antigen ⁴	94.1%	99%
	Serum cryptococcal antigen ⁵	91.4%	83.3%

- 1 Crampin AC, et al. Comparison of two versus three smears in identifying culture-positive tuberculosis patients in a rural African setting with high HIV prevalence. *International Journal of Tuberculosis and Lung Disease*, 2001, 5(11):994–9.
- 2 Wilkinson D et al. Trial-of-antibiotic algorithm for the diagnosis of tuberculosis in a district hospital in a developing country with high HIV prevalence. *International Journal of Tuberculosis and Lung Disease*, 2000, 4(6):513–8.
- 3 Chen S et al. Epidemiology and host- and variety-dependent characteristics of infection due to *Cryptococcus neoformans* in Australia and New Zealand. *Clinical Infectious Diseases*, 2000, 31(2):499–508.
- 4 Antinori S et al. The role of cryptococcal antigen assay in diagnosis and monitoring of cryptococcal meningitis. *Journal of Clinical Microbiology*, 2005, 43(11):5828–9.
- 5 Asawavichienjinda T, Sitthi-Amorn C, Tanyanont V. Serum cryptococcal antigen: diagnostic value in the diagnosis of AIDS-related cryptococcal meningitis. *Journal of the Medical Association of Thailand*, 1999, 82(1):65–71.

5.2 Management of sodium, potassium, and calcium abnormalities

5.2.1 Abnormalities of sodium (Na) concentration

Hypernatraemia (high Na)

Hypernatraemia is an electrolyte disturbance that is defined by an elevated sodium level in the blood. It may occur in patients who are unwell from other causes (such as diarrhoea, diabetic ketoacidosis, or sepsis). The patient may present with symptoms of thirst, fatigue, weakness, or those of the underlying cause. In severe cases, hypernatraemia may present with **emergency signs** such as confusion, coma, or convulsions (see Section 3.5). **Always consider hypernatraemia in each of these situations.**

A history and a clinical evaluation and, in particular, an assessment of the patient's hydration or volume status will help establish the cause of hypernatraemia and guide initial management.

Diagnosis

Serum sodium >145 mmol/litre.

Causes

- Hypernatraemia usually is not caused by an excess of sodium, but rather by a relative deficit of free water in the body. It may occur in the following cases:
 - excessive water loss
 - ◊ gastrointestinal losses – diarrhoea, vomiting
 - ◊ cutaneous losses – high fever, sweating, burns
 - ◊ renal losses – hyperglycaemia (by osmotic diuresis), diabetes insipidus (low ADH secretion that may occur with meningoencephalitis or from drugs such as lithium).
 - insufficient water intake
 - ◊ lack of availability
 - ◊ decreased intake due to decreased level of consciousness.
 - excessive sodium administration
 - ◊ excessive IV normal saline (NS) replacement in hospitalized patients.

Management

- Avoid rapid correction of serum sodium as this can result in cerebral oedema and permanent neurological damage.
- Assess the volume status (hydration) of the patient.
- Calculate volume of fluid to be replaced. In the dehydrated hypernatraemic patient, the volume of water required to correct the deficit can be calculated from the following equation.

$\text{Water deficit (in litres)} = \frac{(\text{serum Na concentration} - 140)}{140} \times 0.5 \times \text{body weight (kg)}$
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E.g. if the serum sodium is 160 mmol in a 70 kg patient, then the total water deficit is $(160-140)/140 \times 0.5 \times 70 = 5$ litres. This volume should be replaced over 48–72 hours. Ongoing losses also need to be factored into fluid replacement.

- Give water orally if the patient is haemodynamically stable and alert, or by nasogastric tube.
- If unable to give water orally, use IV fluid replacement. This is required if the patient is hypovolaemic (increased heart rate, low BP, or postural drop, low JVP, cool peripheries, dry mucosa, decreased skin turgor, or low urine output) or unable to take fluids orally due to decreased level of consciousness. Use normal saline (0.9%) until the patient is haemodynamically stable, then change to 5% dextrose to replace the water deficit. Stop IV fluids when adequate oral intake is established.
- Monitor sodium and other electrolytes twice daily initially, if possible. The serum sodium concentration should be lowered by a maximum of 10 mmol/litre over the first 24 hours.
- Diagnose and treat the underlying cause when possible, and correct other electrolyte abnormalities.

Hyponatraemia (low Na)

Hyponatraemia is an electrolyte disturbance in which the sodium concentration in the blood is lower than normal. It can be a manifestation of a variety of disorders. It is usually only symptomatic when it is severe, or if the onset has been rapid, leading to the development of cerebral oedema. Hyponatraemia may present with nausea, lethargy, confusion, muscle weakness and cramps, and in extreme cases seizures and coma. The signs and symptoms of the underlying cause are likely to be apparent.

Diagnosis

Mild: Na 130–135 mmol/litre
 Moderate: Na 120–129 mmol/litre
 Severe: Na less than 120 mmol/litre

Causes

Hyponatraemia can be caused by many conditions and an assessment of the patient's volume status, used in combination with the calculated osmolality (using the equation below), can indicate the underlying cause and guide management.

$$\text{Osmolality (mmol/l)} = 2 \times (\text{Na} + \text{K}) + \text{urea}/2.8 \text{ mg/dl} + \text{glucose}/18 \text{ mg/dl}$$

(normal range = 280–300 mmol/l)

See summary table below for more details on causes and management. Most causes of hyponatraemia will be associated with a low serum osmolality.

Table: Assessment and management of hyponatraemia according to volume status and serum osmolality

Volume status	Possible causes	Management
Dehydrated or hypovolaemic (increased pulse rate, low BP, or postural drop, low JVP, cool peripheries, dry mucous membranes, decreased skin turgor, low urine output) Classify dehydration according to section 10.7d.2.	Renal losses Diuretics (especially thiazides) Hyperglycaemia (due to osmotic diuresis) Addison's disease Non-renal losses: Gastrointestinal losses (vomiting, diarrhoea, bowel obstruction) Burns	Cautious intravenous hydration using the principles below, and treatment of the underlying cause when possible.
Euvolaemic (normal pulse rate, BP, JVP, peripheries, and urine output)	Serum osmolality <260 mmol/l Syndrome of inappropriate ADH release (SIADH)* Chest disease: TB, pneumonia, abscess CNS disorder: head injury, meningoencephalitis, brain abscess, stroke Malignancy	Treat the underlying cause if possible, and restrict total fluid intake to 50–60% of daily fluid requirement (500–1000 ml on average).
Hypervolaemic (raised JVP, peripheral oedema)	Nephrotic syndrome Cirrhosis Congestive cardiac failure	Treat the underlying cause if possible, and restrict total fluid intake to 50–60% of daily fluid requirement (500–1000 ml on average). May require diuresis.

*Syndrome of inappropriate ADH release (SIADH) is an important cause of low Na but is frequently over-diagnosed; many patients are inappropriately fluid-restricted due to this misdiagnosis. Patients with SIADH are euvolaemic (not dehydrated or oedematous, and not on diuretics). Investigations of a concentrated urine (urine Na >20 mmol/l) in the presence of hyponatraemia (<125 mmol/l) or low plasma osmolality (<260 mmol/kg) confirms this.

Management

Management should be guided by:

- the volume status of the patient
- the likely duration (chronic hyponatraemia is usually symptomatic)
- symptom severity.

Correct Na abnormalities slowly to minimize the risk of permanent neurological deficits or death, which may occur as a consequence of rapid fluid shifts. The increase in serum sodium should be <10 mmol/litre in the first 24 hours and <18 mmol/litre in the first 48 hours.

- In all cases, treat the underlying cause if possible. No further treatment measures are required for asymptomatic or mild hyponatraemia.
- Repeat electrolytes every 12 hours initially to monitor sodium rise, as well as to check for other electrolyte abnormalities.
- In hypovolaemic patients, cautiously hydrate with 0.9% NS to replace the fluid deficit. Use the table in Section 10.7d.2 as a guide to estimate the degree of dehydration. Discontinue fluids when the blood pressure is restored and the patient is euvolaemic.

- In the euvolaemic patient, consider giving a low dose of furosemide (e.g. 40 mg IV) in order to prevent fluid overload while treating the hyponatraemia
- In hypervolaemic patients, treat with 500–1000 ml a day fluid restriction and IV furosemide (40–80 mg). Recheck electrolytes at 4 hours, and then every 6 hours.

In emergency presentations of seizures or coma, the initial correction should be aggressive. Consider using hypertonic saline. If this is not available, use normal saline. Aim for an initial correction of 6 mmol/litre over 4 hours, then a more gradual correction as described above. The rate at which fluid should be given in the initial 4 hours can be calculated from the formula below. The rate of replacement should not exceed 70 mmol/hour.

$\text{Emergency infusion rate (ml/hour)} = 4 \times \text{weight (kg)} / \text{Na concentration of infusion fluid (\%)}$

E.g. the infusion rate of 0.9% normal saline in a 70 kg patient should be $4 \times 70 / 0.9 \approx 300$ ml/hour. However, do not exceed 70 mmol/hour. 1 litre of normal saline (0.9%) contains 154 mmol/l NaCl, i.e. the maximum amount of normal saline that can be given in 1 hour is approximately 450 ml. Hypertonic saline, 3%, has 513 mmol/l of NaCl.

5.2.2 Abnormalities of potassium (K) concentration

Similar to most other electrolyte abnormalities, mild hyperkalaemia and hypokalaemia are often asymptomatic, and are clinically undetectable without a blood test. Severe potassium disturbance may manifest as severe arrhythmia necessitating urgent correction, and may be associated with general lethargy and muscle weakness. Always consider concurrent electrolyte abnormalities.

Hyperkalaemia (high K)

Hyperkalaemia is high serum potassium. It is usually asymptomatic and may be encountered in patients unwell from other causes (diabetic ketoacidosis, septic shock), and is usually diagnosed on routine blood tests or ECGs. Severe hyperkalaemia may be associated with muscle weakness, and can cause sudden serious cardiac arrhythmias and death.

Diagnosis

Mild to moderate: K 5.5–6.5 mmol/l

Severe: K more than 6.5 mmol/l or symptomatic or ECG changes

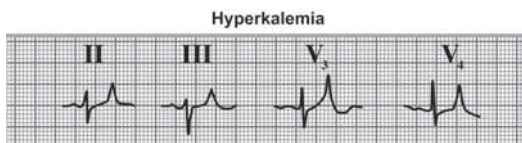
Causes

- Falsely high K reading: haemolysed sample commonly causes an elevated reading as potassium leaks from the cells. Repeat the blood test.
- renal failure
- shock (from any causes)
- diabetic ketoacidosis (hyperglycaemia, insulin deficiency)
- medications: potassium supplements, potassium-sparing diuretics (e.g. spironolactone), ACE inhibitors, non-selective beta-blockers (e.g. atenolol), NSAIDs, heparin
- other: rhabdomyolysis (muscle breakdown), metabolic acidosis, Addison's disease.

Management

- If available, obtain an ECG. Changes occur most markedly in lead V6 and S1. Consider cardiac monitoring or serial ECGs if any of the changes shown below are present.

ECG changes: peaked T waves, prolonged PR interval, small or loss of P waves, widening of the QRS complex progressing to sinusoidal wave, and potentially ventricular tachycardia (VT) or ventricular fibrillation (VF).



- Obtain a repeat sample to check the result, especially if there are no ECG changes.

Treat urgently if ECG changes are present, or if K more than 6.5 mmol/litre.

- Give IV calcium gluconate 1000 mg (10 ml of 10% solution) or calcium chloride 500–1000 mg (5–10 ml of 10% solution) over 2 minutes, to stabilize the cardiac membrane first if ECG changes are present. This can be repeated after 5 minutes if ECG changes persist.
- Give short-acting insulin 10–15 units IV in 50 ml D50 (50% dextrose water) infused over 2 hours, to activate intracellular transfer of K, followed by a dextrose infusion and regular blood glucose monitoring.
- Give salbutamol 10–20 mg by nebulizer or 0.5 mg (500 micrograms) IV. IV administration should be slow, over 15–20 minutes.
 - If these are not available, give salbutamol 1200 micrograms by metered-dose inhaler with spacer (this is 12 puffs).
 - Repeat if necessary, especially if other options are not available.
- Hyperkalaemia associated with severe oliguric renal failure may only be correctable with dialysis, in patients with acute or end-stage renal failure (see Section 11.31), and when the above measures fail. These patients may not have any ECG changes as the increase has been over a long period of time.
- Treat the underlying cause.
- Re-check the serum K to monitor response every 12 hours.
- Repeat all above if necessary.

Note: Most treatment options mentioned here will have little effect in cases of advanced or oliguria renal failure.

Ongoing management and management of mild hyperkalaemia

- Investigate and treat the cause.
- Stop drugs that increase serum K concentration.
- Diuretics, e.g. 20–40 mg furosemide once daily, or a thiazide diuretic, will increase K excretion, and gradually lower K levels over days. Higher doses will be required in renal failure. Except for those who are fluid overloaded, fluid losses should be replaced.

- Kayexelate 15–30 g in 50–100 ml of 20% sorbitol orally or rectally. Be aware of excess Na absorption.
- Avoid potassium-rich foods (e.g. bananas, oranges, mangoes, potatoes, yams, beans, peas, cabbage, and spinach).

Hypokalaemia (low K)

Hypokalaemia is low serum potassium. It is usually asymptomatic but may be symptomatic if the fall in serum potassium is sudden. It may be encountered in patients unwell for other reasons (e.g. diarrhoea, diabetic ketoacidosis, septic shock), and is usually diagnosed on routine blood tests or ECGs. It may also present with muscle weakness and cramps. Severe hypokalaemia may cause sudden serious cardiac arrhythmias and death.

Diagnosis

Mild: K 3.0–3.5 mmol/litre
 Moderate: K 2.5–3.0 mmol/litre
 Severe: K <2.5 mmol/litre, symptoms or ECG changes

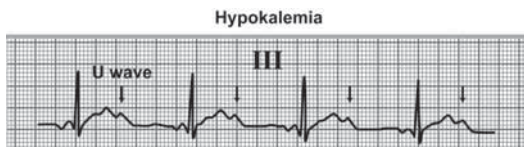
Causes

- gastrointestinal losses (diarrhoea, vomiting)
- medications: diuretics (e.g. furosemide) and chloroquine intoxication
- diabetic ketoacidosis
- other causes: stress response (increased β adrenergic activity), metabolic alkalosis.

Management

- If available, obtain an ECG to help determine the severity

ECG changes: ST depression, flattened or absent T waves, U waves (positive deflection after the T wave), prolonged PR interval, variety of atrial or ventricular arrhythmias.



Mild to moderate hypokalaemia:

- Oral potassium supplements in any preparation (salts, tablet, liquid) should be given at a dose of 10–20 mmol every 6–12 hours. If available, potassium chloride is preferable to citrate or bicarbonate preparations.
- If potassium supplements are not available, encourage the patient to eat potassium-rich foods such as tomatoes, bananas, oranges, melons, mangoes, potatoes, yams, beans, soya beans, peas, cabbage, or spinach.

Severe hypokalaemia:

- Consider cardiac monitoring, especially in patients with ECG abnormalities.
- Use higher doses of oral potassium preparation such as 40 to 60 mmol/l every 6–8 hours.
- In addition, in patients with worrying symptoms, or those who are unable to take oral supplements, give intravenous potassium in saline (dextrose can worsen hypokalaemia initially). **NEVER give a bolus dose of intravenous K as this can cause death.** In most cases, concentrations of 20–40 mmol/l should be used. Caution: more concentrated solutions 100–200 mmol/litre can be used in small volume preparations e.g. 100 ml in patients who are unable to tolerate large infusion volumes. (Particular care should be taken, including ECG monitoring, when concentrated solutions are being infused, as errors in calculating infusion rates may be fatal.)
- The maximal rate of infusion should not exceed 10–20 mmol/hour.
- In all cases, regularly re-check the serum potassium when giving replacements, and look for and treat the underlying cause.

5.2.3 Abnormalities of calcium (Ca) concentration

Hypercalcaemia (high Ca)

Hypercalcaemia is a high serum calcium level. It is most commonly associated with malignancy or parathyroid disease. In mild cases, it is usually asymptomatic; however, when severe, it can present with confusion, coma, or a cardiac arrhythmia. The patient may also present with any of the following symptoms:

- gastrointestinal – abdominal pain, dysphagia, constipation, nausea, vomiting
- renal – dehydration, polyuria, renal stones and renal failure
- neuropsychiatric – anxiety, depression, confusion, seizures, coma
- musculoskeletal – bone pain, weakness.

Diagnosis

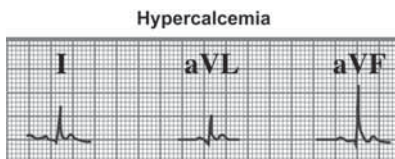
If *serum albumin* can be measured, calculate the more physiologically relevant ionized calcium.

$\text{Ionized calcium} = \text{Ca} + (40 - \text{serum albumin (g/l)}) \times 0.02$
--

Mild:	2.65–3 mmol/litre
Moderate:	3–3.5 mmol/litre and asymptomatic
Severe:	>3.5 mmol/litre or >3.0 and symptomatic or dehydrated

If available, obtain an ECG.

ECG changes: shortened QT interval, widened QRS, flat T waves, AV block, occasional fatal arrhythmias.



Causes

- malignancy
- hyperparathyroidism (primary or tertiary in known renal failure)
- granulomatous disorders – TB, sarcoidosis
- drugs – vitamin D, thiazide diuretics, lithium, indigestion remedies
- other – adrenal failure, hyperthyroidism, immobilization, rhabdomyolysis (muscle breakdown)

Management

Severe hypercalcaemia with CNS symptoms requires urgent treatment.

- Check renal function and electrolytes. Association with hypokalaemia is common and increases the risk of arrhythmias.
- Rehydrate the patient with 0.9% NS at an initial rate of 200–300 ml/hour until urine output >200 ml/hour, then 3–6 litres over 24 hours.
- Determine the rate according to the degree of initial dehydration, medical history (cardiac or renal failure), as well as regular monitoring of urine output, and hydration status (pulse, lying and standing BP, JVP, peripheral perfusion, and oedema). If equipment is available, a urinary catheter may be useful to monitor urine output and fluid balance.
- In a patient with known cardiac or renal impairment, or once the patient is hydrated, use a loop diuretic, e.g. 40 mg furosemide every 4–6 hours with continued IV saline. Electrolytes, especially K and Mg, are likely to fall, and should regularly be checked and supplemented when necessary.
- Steroids (e.g. prednisolone 20–40 mg/day) can be effective in certain etiologies (lymphomas, sarcoidosis, TB, metastases, and vitamin D intoxication).
- Once the patient is stable, aim to investigate and treat the underlying cause.

6. Infection prevention and control

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6. Infection prevention and control

6.1 Principles of hospital infection prevention and control

Infection prevention and control (IPC)^{1,2} are integral to the provision of safe health care. Hospital IPC aims to prevent transmission of communicable diseases including TB,^{3,4} blood-borne and enterically transmitted pathogens, acute respiratory diseases,⁵ as well as to prevent infection during medical procedures (see Section 7 Procedures) or surgery (covered in other sources).

The purpose of IPC includes preventing the transmission of both endemic and epidemic infections. Community-acquired infections can be amplified by transmission within the health facility in the absence of effective IPC practices, with transmission to other patients, visitors, and health workers. These practices are ongoing requirements that apply every day, as well as when there are novel organisms causing an acute respiratory disease or a hemorrhagic fever. This manual for limited-resource settings assumes middle or high TB burden, requiring consistent attention to TB infection control.

Hospital managers should refer to other sources on developing, implementing, and monitoring an IPC programme⁶, training health workers in IPC, providing adequate infection control commodities, assuring a safe blood supply, managing a sterilization section within the hospital,⁷ and improving the infrastructure to make the hospital a safer work environment.

Hospital infrastructure should be arranged and improved as necessary to facilitate hand hygiene, safe waste management, and patient placement. Triage and waiting areas should be well ventilated (open air shelters with a roof are recommended for patient waiting areas), and narrow, poorly ventilated corridors avoided as patient waiting areas. Improving air ventilation⁸ in rooms for patient care includes leaving windows and doors open when possible to maximize cross ventilation. Prioritize

- 1 Core components for infection prevention and control programmes. Report of the Second Meeting of the Informal Network on Infection Prevention and Control in Health Care. WHO, 2008 (WHO/HSE/EPR/2009.1). Available at http://www.who.int/csr/resources/publications/WHO_HSE_EPR_2009_1/en/index.html
- 2 Operations Manual for Delivery of HIV Prevention, Care and Treatment at Primary Health Centres in High-Prevalence, Resource-Constrained Settings. WHO, 2008 (under revision). Available at http://www.who.int/hiv/pub/mai/operations_manual/en/index.html
- 3 WHO policy on TB infection control in health care facilities, congregate settings and households. WHO, 2009. Available at http://whqlibdoc.who.int/publications/2009/9789241598323_eng.pdf
- 4 Implementing the WHO Policy on TB Infection Control in Health-Care Facilities, Congregate Settings and Households. A framework to plan, implement and scale-up TB infection control activities at country, facility and community level. TBCTA, 2010. Available at http://www.stoptb.org/wg/tb_hiv/assets/documents/TBCTImplementationFramework1288971813.pdf
- 5 WHO interim guidance: infection prevention and control in health care for confirmed or suspected cases of pandemic (H1N1) 2009 and influenza-like illnesses. WHO, 16 December 2009 (updated from 29 April 2009 and 25 June 2009 versions). Available at <http://www.who.int/csr/resources/publications/swineflu/swineinflcont/en/index.html>
- 6 Core components for infection prevention and control programmes. WHO, 2008. Available at http://whqlibdoc.who.int/hq/2009/WHO_HSE_EPR_2009.1_eng.pdf
- 7 Sterilization Manual for Health Centers. AMRO-PAHO and USAID, 2009 http://new.paho.org/hq/index.php?option=com_content&task=view&id=2106&Itemid=229&lang=en
- 8 Natural Ventilation for Infection Control in Health-Care Settings. WHO, 2009. Available at http://whqlibdoc.who.int/publications/2009/9789241547857_eng.pdf

IPC recommendations based on assessment of the risk of nosocomial infection in the specific health-care facility and in specific patient care areas.

This Section is aimed at health workers who should refer to the IPC guidelines and use appropriate precautions in their clinical work.

Health worker role in hospital infection prevention and control

- **Ensure a safe working environment.** A safe hospital environment is a high priority for the well-being of staff, patients and visitors. Each health worker should promote a climate of safety to prevent transmission of pathogens in the hospital.
- Standard infection control precautions⁹ should be used, as a minimum, in the care of all patients, staff, and visitors. Standard precautions are meant to reduce the risks of transmission of pathogens from both recognized and unrecognized sources.
- **Assess the risk** of exposure to body substances or contaminated surfaces **BEFORE** any health-care activity. **Make this a routine!** Risk assessment is critical. Assess all health-care activities to determine the level of risk then use appropriate personal protection equipment (PPE) (see Section 6.3).
- Implement source control measures for all persons with respiratory symptoms through promotion of respiratory hygiene and cough etiquette (see Section 6.4).
- Triage, early detection, or suspicion of particular diseases can lead to appropriate seating, hospitalization, and isolation precautions, which can reduce transmission.

Standard precautions for all patients include:^{10,11}

- hand hygiene (see Section 6.2)
- appropriate personal protective equipment (PPE) (see Section 6.3):
 - gloves
 - facial protection (eyes, nose, and mouth)
 - gown
- respiratory hygiene and cough etiquette (see Section 6.4)
- prevention (and management) of injuries from sharp instruments (see Section 6.5)
- environmental cleaning (see Section 6.6)
- appropriate handling of contaminated linens (see Section 6.7)
- waste disposal (see Section 6.8)
- patient care equipment (see Section 6.9).

9 *Infection control standard precautions in health care.* WHO, 2006. Available at http://www.who.int/csr/resources/publications/4EPR_AM2.pdf

10 *WHO interim guidelines on infection prevention and control of epidemic and pandemic-prone acute respiratory diseases in health care.* WHO, 2007. Available at http://www.who.int/csr/resources/publications/swineflu/WHO_CD_EPR_2007_6/en/index.html

11 *Infection control standard precautions in health care.* WHO, 2006. Available at http://www.who.int/csr/resources/publications/4EPR_AM2.pdf

6.2 Hand hygiene¹²

- Ensure availability of hand-washing facilities with clean running water.
- Ensure availability of hand hygiene products (clean water, soap, single-use clean towels, and alcohol-based hand rub). Alcohol-based hand rubs should be made available at every point of care and are the standard of care.
- When to wash hands with soap and running water:
 - when hands are visibly dirty.
- When to use alcohol-based hand rub:
 - when hands appear clean (i.e. are not visibly soiled).

Indications for hand hygiene

- Before and after any direct contact between a health worker and a patient and contact between patients, whether or not gloves are worn. Hands should be washed before gloves are put on.
- Immediately after gloves are removed.
- Before handling an invasive device.
- After touching blood, body fluids, secretions, excretions, non-intact skin, and contaminated items, even if gloves are worn.
- During care, e.g. when moving from a contaminated to a clean body site of the same patient.
- After contact with inanimate objects in the immediate vicinity of the patient.
- Ensure that hands are dry before starting any activity.
- Dry hands with single-use towels.

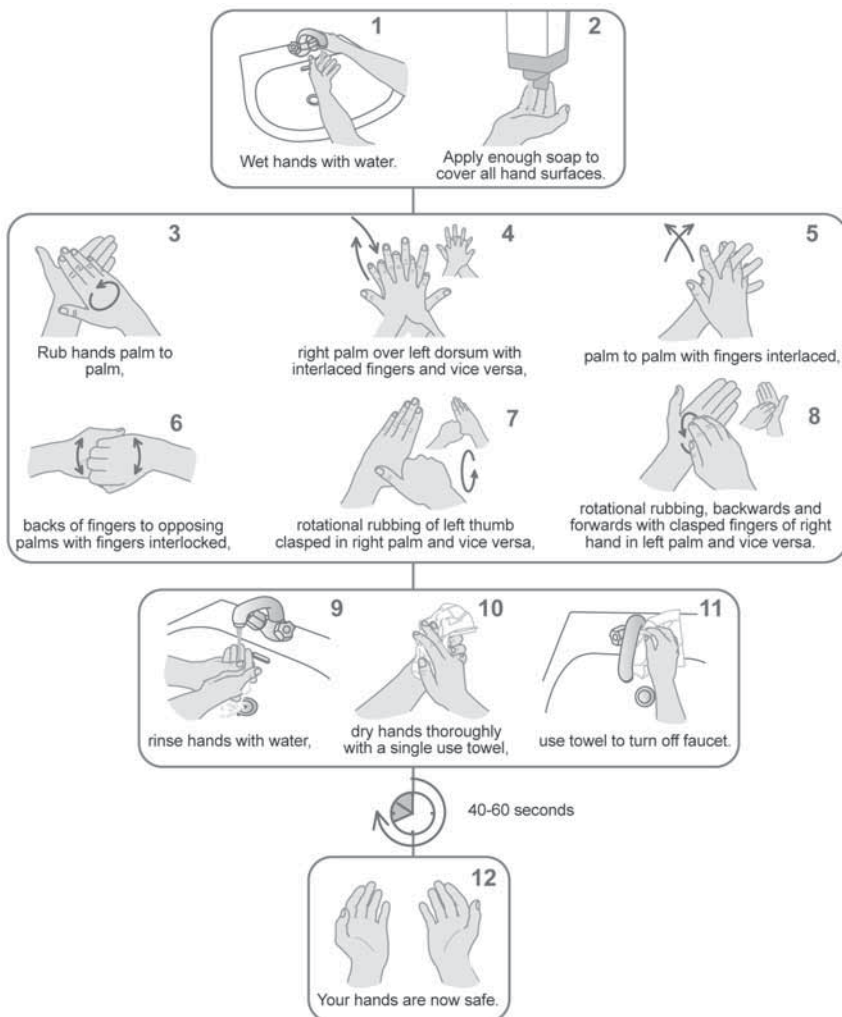
12 WHO *Guidelines on Hand Hygiene in Health Care*. WHO, 2009. Available at http://whqlibdoc.who.int/publications/2009/9789241597906_eng.pdf

Techniques for hand hygiene

Hand washing (40–60 seconds)

- Wet hands and apply soap; rub all surfaces; rinse hands and dry thoroughly with a single use towel; use towel to turn off faucet and dispose of the used towel.

Figure: How to wash the hands with soap and water¹³



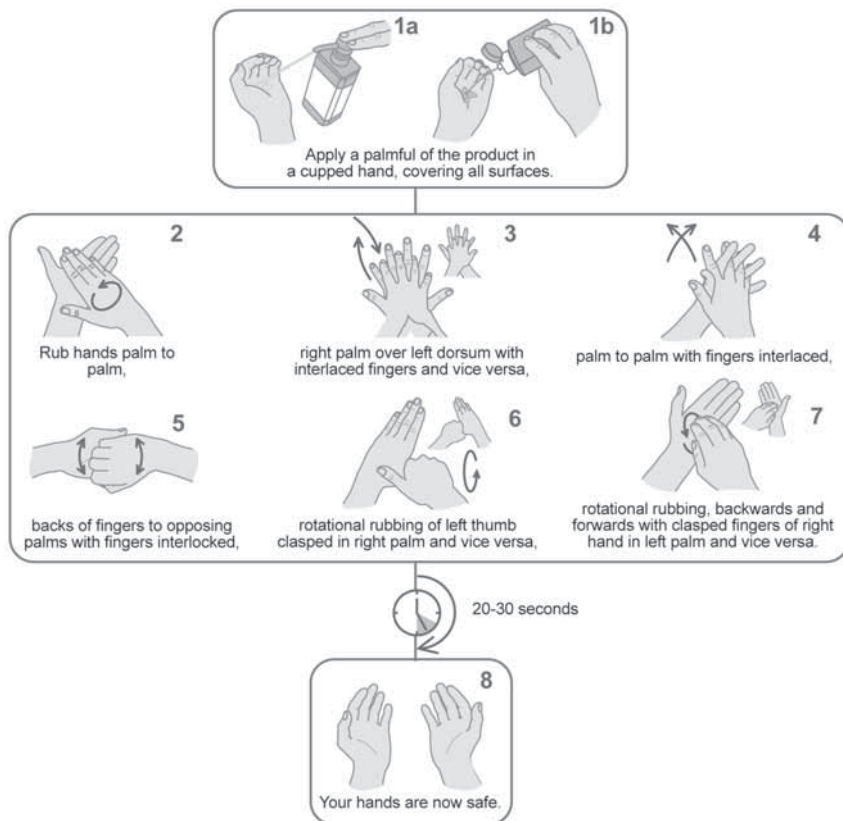
13 WHO Guidelines on Hand Hygiene in Health Care. WHO, 2009. Available at http://whqlibdoc.who.int/hq/2009/WHO_IER_PSP_2009.07_eng.pdf

Techniques for hand hygiene

Hand rubbing (20–30 seconds)

- Apply enough product to cover all areas of the hands; rub hands until dry.

Figure: How to cleanse the hands with an alcohol-based formulation¹⁴



¹⁴ Sterilization Manual for Health Centers. AMRO-PAHO and USAID, 2009. Available at http://new.paho.org/hq/index.php?option=com_content&task=view&id=2106&Itemid=229&lang=en

6.3 Appropriate personal protective equipment (PPE)

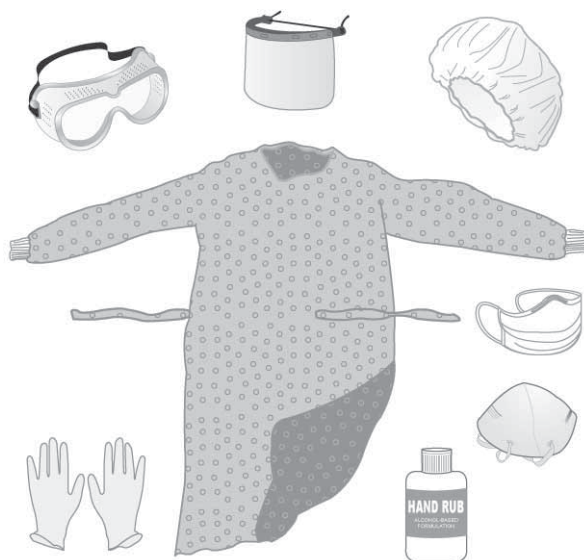
Assess the risk of exposure to body substances or contaminated surfaces **BEFORE** any health-care activity. **Make this a routine!**

- Select PPE based on the assessment of risk:
 - clean, non-sterile gloves
 - clean, non-sterile fluid-resistant gown
 - mask and eye protection or a face shield.
- Ensure that there is a continued supply of PPE.
- Educate and train hospital staff how to wear, remove, and dispose of PPE.

Some PPE is used based on the procedure or type of patient care, no matter what organism (these are part of standard precautions). Additional PPE may need to be added based on the patient's likely diagnosis and suspected pathogen (e.g. if suspect acute respiratory disease of concern, see Section 6.1).

Pathogens differ as to whether they are spread by contact, by large droplets (requiring droplet precautions) or by very small droplet nuclei which can travel more than a meter and stay suspended in the air (requiring airborne precautions).

Figure: Personal protective equipment



PPE to use for any patient according to likely exposure to blood, secretions, non-intact skin

Gloves

- Wear gloves if there is any chance of touching blood, body fluids, secretions, excretions, mucous membranes, or skin, especially skin that is not intact.
- Change between tasks and procedures on the same patient after contact with potentially infectious material, to prevent further contamination.
- Remove after use, before touching non-contaminated items and surfaces, and before going to another patient. Perform hand hygiene immediately after removal.

Facial protection (eyes, nose, and mouth)

- Wear a surgical or procedure mask and eye protection (eye visor, goggles), or a face shield, to protect mucous membranes of the eyes, nose, and mouth during activities that are likely to generate splashes or sprays of blood, body fluids, secretions, or excretions.
- Masks should be used only when it is useful and recommended.

Gown

- Gowns protect the skin and prevent soiling of clothing during activities that are likely to generate splashes or sprays.
- Wear a gown whenever there is any risk of splashes of blood or body fluids.
- If splashing with blood or other body fluids is anticipated and gowns are not fluid-resistant, wear a waterproof apron over the gown.
- Remove soiled gowns as soon as possible, and perform hand hygiene.

Steps to wear PPE

1. Assess risk

2. Select and gather the necessary PPE

3. Put on the gown



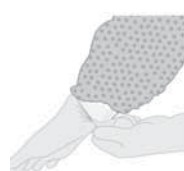
4. Put on the mask



5. Put on eye protection



6. Put on gloves (over cuff)



Steps to remove PPE

1. Peel off gown and gloves and roll inside-out



2. Dispose of safely

3. Perform hand hygiene



4. Remove cap and eye protection (from behind head)



5. Put eye protection in a separate container for reprocessing

6. Remove mask from behind head



7. Perform hand hygiene



6.4 Respiratory hygiene and cough etiquette

- Educate all staff, health workers, patients, and hospital visitors on respiratory hygiene and cough etiquette.
 - Covering mouth and nose when coughing or sneezing.
 - Hand hygiene after contact with respiratory secretions.
 - Spatial separation of persons with acute febrile respiratory symptoms.
- Have tissues available in the waiting area or provide a medical mask.
- When tissues, cloths, or face masks are not available, all staff, health workers, patients, and visitors need to be instructed to lift their arm up and cover their nose and mouth with the inner surface of the arm or forearm when they cough or sneeze.
- Remind all staff, health workers, patients, and visitors to dispose of the tissues and masks in no-touch receptacles and to wash their hands.
- Have posters, at least, in patient waiting areas to remind patients and health workers.

Persons with respiratory symptoms should apply source control measures

- Such persons need to cover their nose and mouth with a tissue or mask when coughing or sneezing, dispose of used tissues and masks appropriately, and perform hand hygiene after coughing or sneezing.

Actions for health-care facilities

- Place patients with acute febrile respiratory symptoms at least 1 metre (3 feet) away from others in common waiting areas.
- Post visual alerts at the entrance to health-care facilities instructing persons with respiratory symptoms to practice respiratory hygiene and cough etiquette.
- Make hand hygiene resources, tissues, and masks available in common areas and areas used for the evaluation of patients with respiratory illnesses.

6.5 Prevention of needle-stick and injuries from other sharp instruments¹⁵

Unsafe injection practices can transmit blood-borne pathogens, including hepatitis B virus (HBV), hepatitis C virus (HCV), and HIV.

Use care when handling, using, cleaning, and disposing of needles, scalpels, and other sharps.

- Do not bend, break, or otherwise manipulate used needles, scalpels, or other sharp instruments.
- Do not recap needles.
- Keep a sharps container nearby when giving injections. Discard single-use needles and syringes immediately after use and directly into the sharps container, without recapping and without passing to another person.
- Close, seal, and send sharps containers for incineration before they are completely full (follow your facility protocol carefully).

Indications for glove use when giving injections	Precautions
Wear non-sterile, well-fitting, single-use gloves: • When there is a likelihood of coming into direct contact with a patient's blood or other potentially infectious materials (e.g. body fluids, moist body substances, and saliva), mucous membranes, and non-intact skin; • When performing venepuncture or venous access injections, because of the potential for blood exposure at the puncture site; • If the health worker's skin is NOT intact or if the patient's skin is NOT intact (e.g. through eczema, cracked or dry skin).	Do not use gloves: • When undertaking routine intradermal, subcutaneous, and intramuscular injections: ◦ if the health worker's skin is intact ◦ if the patient's skin is intact. • Gloves do not provide protection against needle-stick or other puncture wounds caused by sharp objects. • Needles, scalpels and other sharps should be handled with extreme caution.

15 WHO best practices for injections and related procedures toolkit. WHO, 2010. Available at http://whqlibdoc.who.int/publications/2010/9789241599252_eng.pdf

Summary of best practices for injections

DO	DO NOT
• Carry out hand hygiene (use soap and water or alcohol rub), and wash carefully, including wrists and spaces between the fingers, for at least 30 seconds. • Use one pair of non-sterile gloves per procedure or patient. • Use a single-use device for blood sampling and drawing. • Disinfect the skin at the venepuncture site. • Discard the used device (a needle and syringe is a single unit) immediately into a robust sharps container. • If recapping a needle is unavoidable, use the one-hand scoop technique. 1. Leave the needle cap on a flat surface, placed against a firm, upright surface with the cap opening facing towards you. 2. Lift the needle and syringe vertically and guide the tip of the used needle into the cap using only one hand. 3. Once the tip is covered, use the other hand to fix the cap into place. 4. Clean the surface with disinfectant afterwards to avoid leaving any blood. • Seal the sharps container with a tamper-proof lid. • Place laboratory sample tubes in a sturdy rack before injecting into the rubber stopper. • Immediately report any incident or accident linked to a needle or sharps injury, and seek assistance. • Assess for need then start post-exposure prophylaxis (PEP) as soon as possible (see Section 19.6).	• DO NOT forget to clean your hands. • DO NOT use the same pair of gloves for more than one patient. • DO NOT wash gloves for reuse. • DO NOT use a syringe, needle, or lancet for more than one patient. • DO NOT touch the puncture site after disinfecting it. • DO NOT leave an unprotected needle lying outside the sharps container. • DO NOT recap a needle using both hands. • DO NOT overfill or empty sharps from a container. • DO NOT inject into a laboratory tube while holding it with the other hand. • DO NOT delay PEP after exposure to potentially contaminated material. Beyond 72 hours, PEP is NOT effective.

6.6 Environmental cleaning

- Use adequate procedures for the routine cleaning and disinfection of environmental and other frequently touched surfaces.
 - Floors and horizontal work surfaces should be cleaned at least once a day.
 - Cleaning should always be carried out from “clean” areas to “dirty” areas, in order to avoid contaminant transfer.
 - Dry sweeping with a broom should never be done.
 - Rags with dust should not be shaken out and surfaces should not be cleaned with dry rags. Cleaning with a moistened cloth helps to avoid contaminating the air with air-born particles.
- Clean BEFORE you disinfect.
- Change cleaning solutions and equipment frequently, as these items will get contaminated quickly (follow your hospital protocols).

Table: Cleaning, disinfecting, or sterilizing¹⁶

Setting	Manual cleaning with water and detergent	Disinfection (sodium hypochlorite 1% in-use dilution, bleaching powder, alcohol (70%))	Sterilization (steam under pressure, dry heat sterilization, automated chemical)
Floors, work tops	✓		
Spillage – of blood, body fluids, secretions, and excretions	✓	✓	
Commode, toilet seats	✓	✓	
Mops, wash mops	✓		
Dressing trolleys	✓	✓	
Mattress and pillows (always cover with plastic covers)	✓	✓	
Reusable instruments	✓		✓
AMBU bag and mask	✓		✓

¹⁶ Core components for infection prevention and control programmes. WHO, 2008. Available at http://whqlibdoc.who.int/hq/2009/WHO_HSE_EPR_2009.1_eng.pdf

6.7 Linens¹⁷

Handle, transport, and process used linen so as to:

- Prevent skin and mucous membrane exposure and contamination of clothing.
- Avoid transfer of pathogens to other patients or the environment:
 - All used linen and waste should be placed in bags or containers that are able to withstand transportation without being damaged.
 - Any solid matter on soiled linen should be removed and flushed down a toilet.
 - Used linen should be handled carefully to prevent contamination of surrounding surfaces or people.
 - Used linen should be washed according to normal routines.

6.8 Waste disposal

- Ensure safe waste management.
- Treat waste contaminated with blood, body fluids, secretions, and excretions as clinical waste, in accordance with local regulations.
- Human tissue and laboratory waste that is directly associated with specimen processing should be treated as clinical waste.
- Segregate at the point of generation the 4 categories of waste:
 1. sharps
 2. non-sharps infectious waste
 3. non-sharp non-infectious waste
 4. hazardous waste.
- Discard single use items properly.

¹⁷ *Sterilization Manual for Health Centers*. AMRO-PAHO and USAID, 2009. Available at http://new.paho.org/hq/index.php?option=com_content&task=view&id=2106&Itemid=229&lang=en

6.9 Patient care equipment

- Handle equipment soiled with blood, body fluids, secretions, and excretions in a manner that prevents skin and mucous membrane exposure, contamination of clothing, or transfer of pathogens to other patients or the environment.
- Clean, disinfect, sterilize, and reprocess reusable equipment appropriately before use with another patient.

Table: How to set up 3 colour-coded waste containers for most rooms in the hospital (plus a hazardous waste container in the pharmacy and laboratory only)

Waste category	Segregate using colour-coded waste containers	Collect	Dispose
Sharps (needles, scalpels) – infectious or not	YELLOW • Safe sharps container must be: • puncture-proof • covered • closable • upright and stable during use • leak-proof at sides and bottom • clearly labelled for user	• Close lid or cover, seal with tape, and submit for waste pickup when they are no more than $\frac{3}{4}$ full. • Never overfill or force items into these containers. • Collect regularly for disposal.	• Sharps should be disposed of in a sharps pit (a buried drum in small centres or emergency structures, a concrete-lined sealed pit in other settings). • Off-site disposal may be necessary for safe incineration or other safe treatment at the district level (if available) or a private facility in charge of collection and treatment.
Non-sharps infectious waste* (anatomical waste, pathological waste, dressings, used syringes, used single-use gloves)	YELLOW OR RED • Bags or containers • 15–40 litre capacity, with lids	• Containers should be collected, emptied, cleaned, disinfected, and replaced after each intervention (e.g. in an operating or maternity unit) or twice daily. • Bags should not be cleaned and reused but disposed of as sharps infectious waste.	• Non-sharps infectious waste should be buried in a pit fitted with a sealed cover and ventilation pipe for on-site treatment in small health centre settings. • Otherwise, treat on-site or off-site with high-temperature incineration or steam sterilization. • Special arrangements may be needed for disposing of placentas, according to local custom.
Non-sharp, non-infectious waste (paper, packaging)	BLACK • Containers 20–60 litre capacity	• Should be collected, emptied, cleaned and replaced daily. • Alternatively, plastic bags may be used inside the containers for easy removal and disposal.	• May be included in the municipal waste stream or buried in a pit or landfill site. • Non-food and non-medical items may be recycled. • If space is limited, this waste should be incinerated. Ashes and residues should be buried in a pit.

Hazardous waste**	Appropriately labelled containers placed in secure locations.	• These may be stored in a small, labelled container at the pharmacy.	• Follow specific and appropriate treatment protocol and dispose of at the facility or send to a central health facility. • Manage stock of chemicals and pharmaceuticals well to reduce waste quantities and save on purchase costs.
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* Cholera stools, body fluids from other highly infectious diseases.

** Hazardous waste includes some outdated drugs, laboratory reagents, strong disinfectants; radioactive waste, batteries, mercury, etc. Each hazardous waste requires specific treatment and disposal methods based on national regulations.

6.10 Select additional infection control interventions including PPE, based on the risk assessment, epidemiology, or likely pathogen.

Droplet precautions

Additional precautions for infections transmitted by large droplets

A respiratory aerosol of certain infections produces large particles or droplets (>5 µm in diameter) that typically remain suspended in the air for a limited period of time and settle within 1 m (3 feet) of the source.

What to do in addition to standard precautions when such droplet transmission is possible.

- All health workers for all patient care within 1 meter of the patient should wear a medical mask or surgical mask (tight fitting).
- Use single rooms for infectious patients. Otherwise, cohort patients with same suspected etiology. If not possible, place patient beds at least 1 m apart and arranged to keep a distance between patients.

Airborne precautions

Additional precautions for infections transmitted by small droplet nuclei

Smaller particles (small droplet nuclei ≤5 µm in diameter) evaporate quickly; the resulting dried residues settle slowly from the air, and remain suspended in the air for variable lengths of time.

What to do in addition to standard precautions when airborne transmission is possible.

Particulate respirator, e.g. N-95 or similar

- Use adequately ventilated single rooms (≥12 ACH). If single rooms are not possible, cohort patients with the same diagnosis. Airborne precaution rooms can be naturally or mechanically ventilated, with adequate air exchange rate of at least 12 ACH and controlled direction of air flow.

Contact precautions

Additional precautions for infections transmitted by contact

Contact transmission can be direct (direct body surface to body surface contact and physical transfer of micro-organisms) or indirect (e.g. contaminated hands or equipment that carry and transfer the micro-organisms).

What to do in addition to standard precautions

- Gloves and gowns for all patient care.
- Use disposable equipment or dedicate equipment for patient care. If equipment must be shared among patients, clean and disinfect it between each patient use.
- Use single rooms. Otherwise, cohort patients with the same diagnosis. If not possible, place patient beds at least 1 m apart. For pathogens of potential international concern, a single room is more important.

Table: Precautions by suspected organisms – examples

Precaution	Disease or organisms include
Droplet precautions – transmitted by large droplets (in addition to standard precautions)	Any patient with acute respiratory diseases (ARD) transmitted through large droplets. These include human influenza (seasonal, pandemic) and ARD with no pathogen identified (influenza-like illness); pathogens of potential international concern (avian influenza A (H5N1)), SARS*). See Section 6.11.
Airborne precautions – transmitted by small droplet nuclei (in addition to standard precautions)	• Infectious TB – see Section 6.12 • Measles • Varicella
Contact precautions (in addition to standard precautions)	• Adenovirus, para-influenza, RSV • Pathogens of potential international concern (avian influenza A (H5N1)), SARS*) • <i>Vibrio cholera</i>, <i>Shigella</i> species
Standard precautions only (includes all blood-borne pathogens)	There are many known pathogens that do not require additional precautions; however, these still require risk assessment and use of standard precautions. These include common bacterial respiratory infections caused by organisms such as <i>Streptococcus pneumoniae</i> , <i>Haemophilus influenzae</i> , <i>Chlamydia spp.</i> , <i>Mycoplasma pneumoniae</i> . Most blood-borne pathogens including HIV and HBV.

* Note that some organisms require both droplet and contact precautions (in addition to standard precautions).

6.11 Special precautions for acute respiratory diseases (ARDs) that are prone to result in epidemics or pandemics

Separate and fast track patients with or suspected to have ARDs of potential concern

- ARDs of potential concern include SARS-CoV, new influenza viruses causing human infection, and novel ARDs that can cause large-scale outbreaks and outbreaks with high morbidity and mortality.
- Place patients who are coughing or have a suspected ARD of concern in an area separate from other patients and "fast-track" for rapid diagnosis and treatment.
 - They should move to the front of the queue for all services and be assessed promptly.
 - They should wait near an open window or in a comfortable area separate from the general waiting room.
- Accommodate ARD patients at least 1 metre away from other patients.
- For suspected ARDs of concern, prevent contact with contaminated equipment and the environment.
 - Place the patient in a single room or cohort with similarly infected patients.
 - Limit patient unprotected movement and have them wear a mask when moving about.

Table: Precautions for ARDs according to specific clinical settings and procedures¹⁸

Setting or procedure	Infection control measures							
	Hand hygiene	Gloves	Gown	Simple surgical mask	Respirator N95	Eye protection	Respiratory etiquette	Adequately ventilated single room with >12 ACH
Reception (without direct patient contact)							✓	
ER Quick check Physical exam	✓	✓	✓					
Patient waiting area	✓						✓	
General nursing care	✓			✓			✓	
Blood collection	✓	✓		✓			✓	
Nebulization	✓			✓			✓	
Induced sputum	✓	✓	✓		✓	✓		✓

¹⁸ *Infection control strategies for specific procedures in health-care facilities: Quick reference guide.* WHO, 2008. Available at http://whqlibdoc.who.int/hq/2008/WHO_HSE_EPR_2008.2_eng.pdf

Table: Precautions for ARDs according to specific clinical settings and procedures¹⁸

Setting or procedure	Infection control measures							
	Hand hygiene	Gloves	Gown	Simple surgical mask	Respirator N95	Eye protection	Respiratory etiquette	Adequately ventilated single room with >12 ACH
Aerosol-generating procedures associated with pathogen transmission, e.g. intubation or extubation, and manual ventilation, suctioning, autopsy, or surgery involving the use of high-speed devices	✓	✓			✓	✓		✓

6.12 Special precautions for infectious TB patients

- As for Acute respiratory diseases, place patients who are coughing or have suspected TB in an area separate from other patients and have them "fast-tracked" for rapid diagnosis and treatment.
 - They should move to the front of the queue for all services and be assessed promptly.
 - They should wait near an open window or in a comfortable area separate from the general waiting room.
 - "Fast-track" aims to minimize time spent in the hospital for patients suspected of having TB.
- Community-based approaches for the management of TB patients (including MDR-TB) should be prioritized over hospitalization
 - Complement with education of household members and other close contacts on TB infection control.
- Avoid unnecessary admissions of TB patients to health-care facilities.
 - Open doors and windows to use the natural air flow in the hospital.
- On TB wards, the infectious TB patient should wear a medical mask, especially if correct cough etiquette is not observed.
 - The health care workers should wear an N-95 mask when taking care of an infectious TB patient in a close environment.
- Patients with known or suspected drug-resistant TB (DR-TB) should be separated from other patients, including other TB patients.

6.13 Precautions when caring for patients with suspected or confirmed Filovirus (Ebola, Marburg) haemorrhagic fever¹⁹

Careful application of standard precautions should prevent Filovirus haemorrhagic fever transmission.

Current WHO recommendations for direct patient care for known or suspected Filovirus haemorrhagic fever patients

- Restrict all non-essential staff from patient care areas.
- Maintain a log of persons entering the patient's room.
- Limit the number of visitors allowed access to the patient to include only those necessary for the patient's well-being and care, such as a child's parent.
- Ensure that all visitors use PPE according to the facility guidelines. Prior to entering the isolation area, provide all visitors with instructions on using PPE correctly, and instructions for correct hand hygiene practices,
- Do not allow other visitors to enter the care area, and ensure that any visitors wishing to observe the patient do so from an adequate distance from the care area (approximately 15 m).
- Apply infection control precautions to avoid any possible unprotected direct contact with blood and body fluids when providing care to any Filovirus patient, including suspected cases.
 - Perform hand hygiene before and after direct patient care, after any contact with potentially contaminated surfaces, and after removal of PPE. Neglecting to perform hand hygiene after removing PPE will reduce or negate any benefits of the protective equipment.
 - Wear gloves when entering the patient care area.
 - Wear a disposable, impermeable gown to cover clothing and exposed skin. Wear a waterproof apron over any permeable gown or when undertaking any strenuous activity (e.g. carrying a patient).
 - Wear facial protection to prevent splashes to the nose, mouth, and eyes. Facial protection can be achieved by means of (1) medical mask and eye protection (eye visor or goggles), or (2) with a face shield.
- Before exiting the isolation area of a patient with suspected Filovirus infection, carefully remove and dispose of protective equipment.
- When removing protective equipment, be careful to avoid any contact between the soiled items (e.g. gloves, gowns) and any area of the face (eyes, nose, or mouth).
- Ensure that clinical and non-clinical personnel are assigned exclusively to Filovirus patient care areas and that members of staff do not move freely between the isolation areas and other clinical areas during the outbreak.
- Limit the use of needles and other sharp objects as much as possible.
- Limit the use of phlebotomy and laboratory testing to the minimum necessary for essential diagnostic evaluation and patient care.

See Section 19 for TB and HIV prevention and care services for health workers.

¹⁹ *Interim Infection Control Recommendations for Care of Patients with Suspected or Confirmed Filovirus (Ebola, Marburg) Haemorrhagic Fever.* WHO, 2008. Available at http://www.who.int/csr/bioriskreduction/filovirus_infection_control/en/